Saturated & Unsaturated Compounds

Exam code: 0620 & 0971

Download PDF

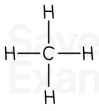
Written by: Alexandra Brennan
Reviewed by: Richard Boole
Updated on 23 January 2026

Saturated & unsaturated Compounds

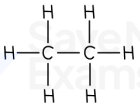
What are saturated compounds?

- **Saturated compounds** are molecules with only **single carbon-carbon bonds** (C-C)
  - Each carbon atom has 4 single bonds to other atoms
- **Alkanes** are examples of saturated compounds
- Alkanes have the general formula $\mathbf{C_nH_{2n+2}}$
- The first three alkanes are:

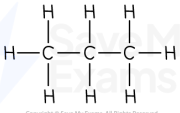
1. Methane, CH_4



2. Ethane, C_2H_6



3. Propane, C_3H_8



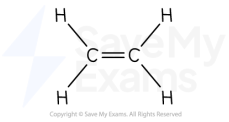
What are unsaturated compounds?

- **Unsaturated compounds** are molecules with one or more carbon-carbon **double bonds** (C=C)
  - Each carbon atom has 4 bonds to other atoms
  - But, one bond is a double bond

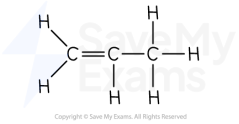
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

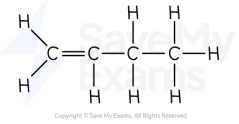
1. Ethene, C_2H_4



2. Propene, C_3H_6



3. Butene, C_4H_8



Examiner Tips and Tricks

Remember:

Saturated compounds have **S**ingle bonds only.

Unsaturated compounds have do**U**ble bonds

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Test yourself

Flashcards

Was this revision note helpful? Yes No

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.